

Back propagation through time

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1 Derivation

The recurrent neural network state is defined by the following equations:

The state at time t is

$$s_t = f(s_t, x_t) = f_1(W^{sx}x_t + W^{ss}s_{t-1} + W_0^{ss})$$

The prediction is given by

$$p_t = g(s_t) = f_2(W^os_t + W_0^o)$$

$$x_t : l \times 1$$

$$s_t : m \times 1$$

$$y_t : v \times 1$$

The weights matrices are of size

$$W^{sx} : m \times l$$

$$W^{ss} : m \times m$$

$$W_0^{ss} : m \times l$$

$$W^o : v \times m$$

$$W_0^o : v \times 1$$

we also define

$$z_t^1 = W^{sx}x_t + W^{ss}s_{t-1} + W_0^{ss}$$

$$z_t^2 = W^os_t + W_0^o$$

such that we can concisly write

$$s_t = f_1(z_t^1)$$

$$p_t = f_2(z_t^2)$$

The backward pass for both W^{ss} and W^{sx} involves computing

$$\frac{dL_{seq}}{dW} = \sum_{u=1}^n \frac{dL_u}{dW}$$

where $L_u = L_{elt}(p_u, y_u)$

The since W effects L_{elt} through all the states we have

$$\begin{aligned} dL_{elt} &= \sum_{t=1}^n \frac{\partial L_u}{\partial s_t} ds_t \\ \frac{dL_{seq}}{dW} &= \sum_{u=1}^n \sum_{t=1}^n \frac{\partial L_{elt}}{\partial s_t} \frac{ds_t}{dW} = \sum_{u=1}^n \sum_{t=1}^n \frac{\partial L_{elt}}{\partial s_t} \frac{\partial s_t}{\partial W} \end{aligned}$$

Since s_t effects only the L_u at steps $u \geq t$ we have

$$\begin{aligned} &= \sum_{u=t}^n \sum_{t=1}^n \frac{\partial L_u}{\partial s_t} \frac{\partial s_t}{\partial W} \\ &= \sum_{t=1}^n \frac{\partial s_t}{\partial W} \sum_{u=t}^n \frac{\partial L_u}{\partial s_t} \\ &= \sum_{t=1}^n \frac{\partial s_t}{\partial W} \left[\frac{\partial L_u}{\partial s_t} + \sum_{u=t+1}^n \frac{\partial L_u}{\partial s_t} \right] \end{aligned}$$

we define the

$$\delta^{s_t} = \sum_{u=t+1}^n \frac{\partial L_u}{\partial s_t}$$

which is the dependence of the loss at times after t on the state at time t .

$$\frac{dL}{dW} = \sum_{t=1}^n \frac{\partial s_t}{\partial W} \left[\frac{\partial L_u}{\partial s_t} + \delta^{s_t} \right]$$

Defining the future loss

$$F_t = \sum_{u=t+1}^n L_u = \sum_{u=t+1}^n L_{elt}(p_u, y_u)$$

$$\begin{aligned} \delta^{s_{t-1}} &= \sum_{u=t}^n \frac{\partial L_u}{\partial s_{t-1}} = \frac{\partial s_t}{\partial s_{t-1}} \left[\frac{\partial L_u}{\partial s_t} + \sum_{u=t+1}^n \frac{\partial L_u}{\partial s_t} \right] \\ &= \frac{\partial s_t}{\partial s_{t-1}} \left[\frac{\partial L_t}{\partial s_t} + \sum_{u=t+1}^n \frac{\partial L_u}{\partial s_t} \right] \end{aligned}$$

$$\delta^{s_{t-1}} = \frac{\partial s_t}{\partial s_{t-1}} \left[\frac{\partial L_t}{\partial s_t} + \delta^{s_t} \right]$$

note that $s_t = f_1(W^{sx}x_t + W^{ss}s_{t-1} + W_0^{ss})$

In more detail we have that δ^{s_t} and $\frac{\partial L_u}{\partial s_t}$ are vector of size $m \times 1$, therefore the element k is given by

$$\delta_k^{s_{t-1}} = \sum_{u=t}^n \frac{\partial L_u}{\partial s_{t-1}^k} = \sum_j \frac{\partial s_t^j}{\partial s_{t-1}^k} \left[\frac{\partial L_t}{\partial s_t^j} + \sum_{u=t+1}^n \frac{\partial L_u}{\partial s_t^j} \right]$$

$$\frac{\partial s_t^j}{\partial s_{t-1}^k} = W_{jk}^{ss} \frac{\partial f}{\partial z_t^{1,j}}$$

Therefore we see that $\frac{\partial s_t}{\partial s_{t-1}}$ is a $m \times m$ matrix which k, j element is given by $(W^{ss})_{kj}^T \frac{\partial f}{\partial z_t^{1,j}}$

$$\rightarrow \frac{\partial s_t}{\partial s_{t-1}} = (W^{ss})^T * \frac{\partial f}{\partial z_t^1}$$

where $*$ is the Hadamard product (element by element product)

This can be see since $s_t : m \times 1$, therefore $\frac{\partial s_t}{\partial s_{t-1}} : m \times m$, $\frac{\partial f}{\partial z_t^1} : m \times 1$ and $W^{ss} : m \times m$ (the transpose is needed to get the order correct.)

In addition

$$\begin{aligned} \frac{\partial L_u}{\partial s_t} &= \frac{\partial z_t}{\partial s_t} \frac{\partial L_u}{\partial z_t} \\ &= (W^o)^T * \frac{\partial L_u}{\partial z_t^2} \end{aligned}$$

This fits, since $W^o : v \times m$ and $\frac{\partial L_u}{\partial z_t^2} : v \times 1$

overall

$$\delta^{s_t} = (W^{ss})^T * \frac{\partial f}{\partial z_t^1} \left[(W^o)^T * \frac{\partial L_u}{\partial z_t^2} + \delta^{s_t} \right]$$

$$\begin{aligned} \frac{L_{seq}}{dW^{ss}} &= \sum_{t=1}^n \frac{\partial s_t}{\partial W^{ss}} \sum_{u=t}^n \frac{\partial L_u}{\partial s_t} \\ &= \sum_{t=1}^n \frac{\partial s_t}{\partial W^{ss}} \frac{\partial F_u}{\partial s_t} \\ &= \sum_{t=1}^n \frac{\partial z_t^1}{\partial W^{ss}} \frac{\partial s_t}{\partial z_t^1} \frac{\partial F_u}{\partial s_t} \end{aligned}$$

To understand the exact matrix expression we look at the specific index

$$\left[\frac{L_{seq}}{dW^{ss}} \right]_{jk} = \frac{L_{seq}}{dW_{jk}^{ss}} = \sum_p \sum_{t=1}^n \frac{\partial s_t^p}{\partial W_{jk}^{ss}} \frac{\partial F_u}{\partial s_t^p} = \sum_p \sum_{t=1}^n \frac{\partial s_t^p}{\partial W_{jk}^{ss}} \delta_p^{s_t}$$

since the relation between s_t and z_t is an elementwise operation (the f_1 mapping) we have

$$\begin{aligned}
&= \sum_p \sum_{t=1}^n \frac{\partial z_t^{1,p}}{\partial W_{jk}^{ss}} \frac{\partial s_t^p}{\partial z_t^{1,p}} \delta_p^{s_t} \\
&= \sum_p \sum_{t=1}^n s_{t-1}^p \delta_{jp} f'(z_t^{1,p}) \delta_p^{s_t} \\
&= \sum_{t=1}^n \left(\frac{\partial f_1}{\partial z_t^1} \right)^T s_{t-1} * \frac{\partial F_u}{\partial s_t}
\end{aligned}$$

$$\frac{\partial f_1}{\partial z_t^1} : m \times 1, s_t : m \times 1$$

But I'm not sure regarding the order of the last step, write it in terms of indicies and work it out...

2 Homework solutions (week 11)

Q4

We consider the following NN

$$s_1 = W^{ss} s_0 + v_1$$

$$v_1 = \sigma(w_1)$$

$$w_1 = CNN(x_1)$$

$$p_1 = W^o s_1$$

$$s_2 = W^{ss} s_1 + v_2$$

$$v_2 = \sigma(w_2)$$

$$w_2 = CNN(x_2)$$

$$p_2 = W^o s_2$$

$$L(p, y) = \frac{1}{2T} \sum_{t=1}^T (p_t - y_t)^2$$

W^{ss} and W^o is a scalar

$$\begin{aligned}
\frac{\partial L}{\partial v_2} &= \frac{1}{2} (p_2 - y_2) \frac{\partial}{\partial v_2} p_2 = \frac{1}{2} (p_2 - y_2) \frac{\partial}{\partial v_2} W^o s_2 \\
&= \frac{1}{2} (p_2 - y_2) \frac{\partial}{\partial v_2} (W^o (W^{ss} s_1 + v_2)) = \frac{1}{2} (p_2 - y_2) W^o
\end{aligned}$$

$$\begin{aligned}
\frac{\partial L}{\partial v_1} &= \frac{1}{2} (p_2 - y_2) \frac{\partial}{\partial v_1} W^o s_2 + \frac{1}{2} (p_1 - y_1) \frac{\partial}{\partial v_1} p_1 \\
&= \frac{1}{2} (p_2 - y_2) \frac{\partial}{\partial v_1} (W^{ss} s_1 + v_2) + \frac{1}{2} (p_1 - y_1) W^o \\
&= \frac{1}{2} (p_2 - y_2) W^o W^{ss} \frac{\partial}{\partial v_1} s_1 + \frac{1}{2} (p_1 - y_1) W^o \\
&= \frac{1}{2} [(p_2 - y_2) W^o W^{ss} + (p_1 - y_1) W^o]
\end{aligned}$$

Q5

51.A

$$\frac{\partial L}{\partial s_t} = \frac{\partial z_t^2}{\partial s_t} \frac{\partial L}{\partial z_t^2} = (W^o)^T (p_t - y_t)$$

51.B

$$\begin{aligned}
&\frac{\partial}{\partial s_t} \left[L(y_t, p_t) + \sum_{l=t+1}^n L(y_l, p_l) \right] \\
&= \frac{\partial}{\partial s_t} L(y_t, p_t) + \frac{\partial}{\partial s_t} F_t
\end{aligned}$$

51.C

$$\begin{aligned}
&\frac{\partial}{\partial z_t^1} \left[L(y_t, p_t) + \sum_{l=t+1}^n L(y_l, p_l) \right] \\
&= \frac{\partial s_t}{\partial z_t^1} \left[\frac{\partial}{\partial s_t} L(y_t, p_t) + \frac{\partial F_t}{\partial s_t} \right] \\
&= f_1'(z_t^1) * \left[\frac{\partial}{\partial s_t} L(y_t, p_t) + \frac{\partial F_t}{\partial s_t} \right]
\end{aligned}$$

51.D

$$\begin{aligned}
\delta^{s_{t-1}} &= \sum_{u=t}^n \frac{\partial L_u}{\partial s_{t-1}} = \frac{\partial z_t}{\partial s_{t-1}} \left[\frac{\partial L_u}{\partial z_t} + \sum_{u=t+1}^n \frac{\partial L_u}{\partial z_t} \right] \\
&= (W^{ss})^T \cdot \left[\frac{\partial L_u}{\partial z_t} + \sum_{u=t+1}^n \frac{\partial L_u}{\partial z_t} \right]
\end{aligned}$$

52.A

We have

$$\frac{dF}{ds_t} = \frac{\partial}{\partial s_t} \left[L_t + \sum_{l=t+1}^n L_l \right]$$

$$\frac{dF}{dz_t^1} = \frac{\partial}{\partial z_t^1} \left[L_t + \sum_{l=t+1}^n L_l \right]$$

$$\frac{dL}{dz_t^2}$$

$$\frac{d}{dW^o} L(y_t, p_t) = \frac{\partial L}{\partial z_t^2} \frac{\partial z_t^2}{\partial W^o}$$

$$= \frac{\partial L}{\partial z_t^2} s_t^T$$

52.B

$$\frac{d}{dW^o} L(y_t, p_t) = \frac{\partial L}{\partial z_t^2} \frac{\partial z_t^2}{\partial W_0^o} = \frac{\partial L}{\partial z_t^2} \frac{\partial z_t^2}{\partial W_0^o}$$

52.C

$$\frac{d}{dW^{ss}} \left[\sum_{l=t}^n L_l \right] = \frac{\partial z_t^1}{\partial W^{ss}} \frac{\partial}{\partial z_t^1} \left[\sum_{l=t}^n L_l \right]$$

$$= \left[\frac{\partial}{\partial z_t^1} \sum_{l=t}^n L_l \right] \cdot s_{t-1}^T$$

52.D

$$\frac{d}{dW_0^{ss}} \left[\sum_{l=t}^n L_l \right] = \frac{\partial z_t^1}{\partial W^{ss}} \frac{\partial}{\partial z_t^1} \left[\sum_{l=t}^n L_l \right] = \frac{\partial}{\partial z_t^1} \left[\sum_{l=t}^n L_l \right]$$

52.E

$$\frac{d}{dW^{sx}} \left[\sum_{l=t}^n L_l \right] = \frac{\partial z_t^1}{\partial W^{sx}} \frac{\partial}{\partial z_t^1} \left[\sum_{l=t}^n L_l \right] = \left[\frac{\partial}{\partial z_t^1} \sum_{l=t}^n L_l \right] \cdot x_t^T$$